

Crop Stress Early Warning That Fires Before NDVI Moves

BUCKET 10 - CHANGE DETECTION AND EARLY WARNING

For EU arable farmers and CAP/insurance monitoring bodies who today only learn a field is stressed once NDVI has already dropped (i.e. after yield is lost), a self-supervised anomaly detector fusing Sentinel-1 SAR and Sentinel-2 optical time series learns each parcel's normal phenological trajectory and flags divergence days before the NDVI-threshold rule fires, with the technical core being a fused-modality temporal autoencoder scored against a measured lead-time-in-days metric rather than a classification accuracy number.

VIABLE

The idea passes all six hard constraints and has a genuinely defensible novelty delta versus the named STS-NET baseline, but the single biggest open question - whether any real, non-circular ground-truth stress-event record can be obtained - remains unresolved after this search and must be the very first thing the student verifies, alongside an early empirical check of whether SAR actually adds lead time at all.

Confidence: Medium. The arXiv/OpenAlex/Crossref/dblp searches and WebSearch dataset-access checks were thorough and internally consistent (STS-NET's abstract, the ASAP/Meroni baselines, and the Huo et al. counter-evidence all corroborate each other), but this pass did not directly query the Copernicus STAC catalogue for actual local revisit statistics on a specific candidate AOI, nor did it contact any EU national crop-monitoring body (e.g. a Sen4CAP partner) to confirm dated stress-event data availability. Confidence would rise materially with either of those two direct checks.

Hard constraint check

Constraint	Verdict	Reasoning
H1 · Novelty	PASS	genuine two-part delta (new SAR+optical data regime, new lead-time-in-days evaluation objective) versus the named STS-NET baseline, not a mere region re-application.
H2 · Not generic	PASS	not any of the 8 banned patterns; explicitly positioned against, not as, a plain NDVI viewer.
H3 · Imagery is primary input	PASS	Sentinel-1 SAR and Sentinel-2 optical time series are the sole primary inputs; sensor, resolution, revisit, and licence are documented above.
H4 · Geography & user	CAVEAT	EU arable farmers/insurers is the better-evidenced user segment given available ground-truth infrastructure; the Singapore/Malaysia framing is weaker and needs explicit justification or should be dropped in favour of EU.
H5 · Compute & budget	PASS	small model architecture, small AOI dataset (15-30GB), all free-tier data and compute, comfortably under the 24GB/12h/RM200/200GB ceilings.
H6 · Pitch quality	PASS	'an alert that fires before NDVI moves, measured in days, beating what insurers use today' is a clear 60-second pitch with a concrete, live, dated demo.

The problem

The dominant operational practice for crop stress monitoring, used by farmers, agronomists, and index-insurance schemes, is to watch NDVI (or similar optical vegetation indices) and raise an alert when it drops below a threshold or deviates from its historical seasonal norm (as in the EC JRC ASAP/MARS system and near-real-time MODIS NDVI anomaly monitoring). This is fundamentally a lagging indicator: canopy greenness only changes after physiological damage (cell/tissue-level water or nitrogen deficit, early disease) has already occurred, so by the time the NDVI-threshold rule fires, some yield loss is already locked in. Farmers and insurers have no widely deployed method that is dated, alerts before the optical signal moves, and is validated on how many days of lead time it actually buys, as opposed to how accurately it classifies stress after the fact.

Proposed technical approach

A self-supervised anomaly detector operating on joint Sentinel-1 (VV/VH backscatter) and Sentinel-2 (NDVI/NDRE/GNDVI-derived optical time series) per-parcel time series. Architecture: a 3D convolutional or ConvLSTM-based autoencoder (following the general recipe of STS-NET's 3D-CAE but extended to two input streams) that learns to reconstruct/predict each parcel's expected multi-band trajectory conditioned on its own phenological history and crop-type/growth-stage context; the reconstruction error (or a learned normality score) at each new acquisition is the anomaly signal. Training objective: self-supervised reconstruction or next-step prediction loss over historical, presumed-normal seasons only (no stress labels needed for training). Inference/output: for each parcel and each new Sentinel-1/2 acquisition, a dated anomaly score; an alert fires the first time the score crosses a threshold calibrated on a held-out normal-season set. Evaluation: lead-time-in-days = date(NDVI-threshold-rule alert) minus date(fused-model alert), computed per parcel, compared against a named NDVI-only anomaly baseline reproducing the ASAP/Meroni et al. near-real-time NDVI anomaly logic. A SAR-only vs. optical-only vs. fused ablation isolates whether the SAR channel is actually contributing lead time (this ablation is a mandatory early check, see risks).

Novelty

WHAT THE DELTA ACTUALLY IS

STS-NET (arXiv:2607.18791, 2026) establishes that self-supervised crop-stress detection from satellite image time series is viable, using a 3D convolutional autoencoder over four optical vegetation indices derived from commercial PlanetScope imagery, evaluated by classification precision on sugarcane stress types. It explicitly does not use SAR and does not measure or report anything resembling lead time - its headline numbers are precision percentages, not days-early. The candidate's delta is two orthogonal, defensible changes, not a region swap: (1) a new data regime - free-tier Sentinel-1 SAR fused with Sentinel-2 optical, exploiting a physically distinct sensing channel (radar backscatter sensitive to canopy structure/water content) that the cited prior art never uses; (2) a new evaluation objective - the headline number is lead-time-in-days measured against a named, currently-deployed NDVI-anomaly practice (ASAP/Meroni), not a classification accuracy score. Neither change is present in the cited baseline, and 'applied a self-supervised detector to Sentinel data' would not by itself pass H1 without the lead-time framing, which is the actual technical claim.

PRIOR ART THAT THREATENS IT

Huo, Persson & Lindberg (2021, RSE, DOI 10.1016/j.rse.2020.112240) already fused Sentinel-1 and Sentinel-2 to detect pre-visual forest stress (European spruce bark beetle) and found that Sentinel-1 backscatter contributed only marginally to classification accuracy beyond the red/SWIR optical bands - this is a genuine, closely analogous result that threatens the specific claim that 'SAR is the key early-signal channel' for crops; the domain differs (forest pest vs. crop water/nitrogen stress) so it is not dispositive,

but it means the SAR contribution must be empirically demonstrated, not assumed. Le Page et al. (2020, RS MDPI, DOI 10.3390/rs12101621) already showed Sentinel-1 soil-moisture-derived signals can detect irrigation events at plot scale, supporting the premise that SAR senses water-status changes early, but this was not assembled into a self-supervised anomaly detector with a lead-time metric. No 2024-2026 paper was found (arXiv, OpenAlex, Crossref, dblp all searched) that combines self-supervised S1+S2 fusion anomaly detection with an explicit lead-time-in-days metric benchmarked against a named NDVI-threshold baseline; this specific combination appears to remain open, though the individual ingredients (fusion, self-supervision, pre-visual stress signal) are each separately precedented.

GENERIC-PATTERN CHECK

Confirmed **NOT** a banned pattern: not a chatbot/RAG assistant, not a dashboard/monitoring portal, not leaf-photo disease classification, explicitly **NOT** a plain NDVI viewer (the whole point is to beat NDVI), not tabular yield prediction from weather/soil CSVs, not a crop-type-classification benchmark rerun, not an API-wrapper mobile app. Could the one-sentence summary describe 50 other FYPs? No - most 'satellite crop monitoring' student projects report classification accuracy after the fact; committing to a measured days-of-lead-time headline metric against a named operational baseline is a specific, falsifiable technical claim that is uncommon in typical FYP framing.

Sensors and imagery

Sensor	Resolution	Revisit (days)	Bands	Licence
Sentinel-1	10 (resampled from ~5x20 native) m	6-12 (varies by orbit/region; ~6 days over parts of EU with dual-satellite coverage prior to S1B loss, currently single-satellite ~12 days pending S1C/D)	VV, VH backscatter	Copernicus free and open data policy, no restriction on commercial/non-commercial use
Sentinel-2	10 (visible/NIR), 20 (red-edge/SWIR) m	5 (combined S2A+B, cloud-dependent effective revisit often worse)	Red, NIR, red-edge (B5/B6/B7), SWIR for NDVI/NDRE/GNDVI	Copernicus free and open data policy

Both sensors fused at the parcel-time-series level; imagery is the sole primary input, no weather/soil CSV fusion in the core model

Yes - Sentinel-1 SAR and Sentinel-2 optical time series are the only model inputs; parcel geometry (from AI4Boundaries/LPIS) is used only to define the extraction unit, not as a predictive feature.

Datasets

COPERNICUS DATA SPACE ECOSYSTEM (SENTINEL-1 GRD + SENTINEL-2 L2A)

Access route	dataspace.copernicus.eu via STAC/openEO/Sentinel Hub API and direct product download
Auth required	Free registration only, no payment
Licence	Copernicus free and open data policy, no commercial-use restriction, quotas apply for bulk/large-scale processing
Approx. size	5-20 for a single-region pilot AOI over one growing season (full-tile downloads are larger; a small AOI clipped via STAC/openEO keeps this well under 20GB)
Accessible from MY	Yes, verified globally accessible, no geoblocking
How verified	WebSearch confirmed dataspace.copernicus.eu live in 2026 with STAC/openEO/Sentinel Hub APIs and free-of-charge access with quotas for large-scale use (Sentinel Online news item and dataspace.copernicus.eu itself)

AI4BOUNDARIES (EU PARCEL BOUNDARIES + SENTINEL-2 MONTHLY COMPOSITES)

Access route	JRC Open Data Catalogue, http://data.europa.eu/89h/0e79ce5d-e4c8-4721-8773-59a4acf2c9c9 , plus github.com/waldnerf/ai4boundaries helper library
Auth required	None
Licence	Open (JRC/EU open data)
Approx. size	~5-10 (7,831 samples of 256x256px S2 plus 1m orthophotos for a subset)
Accessible from MY	Yes
How verified	OpenAlex record (d'Andrimont et al. 2023, essd-15-317-2023) plus WebSearch confirming live JRC catalogue and GitHub helper repo

STS-NET'S BSPT (BARNALA SPATIAL-TEMPORAL) DATASET

Access route	Not publicly released; author-built private dataset used only inside the STS-NET paper
Auth required	N/A - not accessible
Licence	Unknown/not stated
Approx. size	Unknown
Accessible from MY	No - not publicly available
How verified	Read the STS-NET abstract via arXiv (2607.18791); no dataset release link found in abstract or accompanying search

EC JRC ASAP / MARS BULLETINS (ADMIN-UNIT-LEVEL NDVI/RAINFALL ANOMALY OUTPUTS)

Access route	mars.jrc.ec.europa.eu/asap
Auth required	None for viewing; bulk data access terms not fully verified
Licence	Free to view, reuse terms [uncertain]
Approx. size	Small (bulletins, not raw imagery)
Accessible from MY	Yes
How verified	OpenAlex record for Rembold et al. 2019 (ASAP paper) describing the public website; not independently re-fetched via WebFetch in this pass [uncertain]

TOTAL DATA VOLUME

Approximately 15-30 GB for a minimum viable pilot (one or two EU regions, one growing season, Sentinel-1+2 time series clipped to a modest AOI plus AI4Boundaries parcel geometries) - well under the ~200GB practical H5 ceiling.

LABELS AND GROUND TRUTH

The model itself is trained self-supervised (reconstruction/prediction error against each parcel's own historical normal trajectory) and needs no stress labels for training. The hard dependency is on the EVALUATION side: measuring 'lead time' credibly requires a dated, independent record of when real crop stress actually began. A systematic search (arXiv, OpenAlex, Crossref, WebSearch) did not surface a public, parcel-level, dated crop-stress-onset dataset for either the EU or Southeast Asia at the needed granularity - STS-NET's own BSPT ground-truth dataset is private and covers one 2.5-acre plot in India. The workable substitute is a self-referential proxy: define the comparison event as the date the NDVI-threshold rule itself would have fired (replicating the ASAP/Meroni logic), and measure the fused model's lead time relative to that, rather than to an independent physiological ground truth. This is defensible as a direct head-to-head against the named baseline practice, but it does not, by itself, prove the alert corresponds to real agronomic stress - a small set of field-verified events (via a local agronomist contact or a well-documented historical drought/flood year) is needed to sanity-check that the earlier alerts are not simply noise-driven false positives. **[uncertain: whether any usable dated ground-truth stress record exists beyond this self-referential proxy]**

Baselines to beat

Paper	Reported metric	Dataset
STS-NET: Spatio-Temporal Stress Network for Self-Supervised Crop Stress Detection using Satellite Image Time Series arXiv preprint 2026 · arXiv: 2607.18791	Precision 97.98% (water stress), 85.08% (nitrogen stress), 83.47% (combined stress)	BSPT (private, Barnala) for training; a 2.5-acre sugarcane test plot in Lakhimpur-Kheri, India for evaluation, using PlanetScope-derived NDVI/ GNDVI/RECI/ NDRE

Near real-time vegetation anomaly detection with MODIS NDVI: Timeliness vs. accuracy and effect of anomaly computation options Remote Sensing of Environment 2019 · 10.1016/j.rse.2018.11.041	Quantifies estimation error and delay of near-real-time NDVI anomaly indicators (standard scores, non-exceedance probability, vegetation condition index)	1km MODIS NDVI, global
ASAP: A new global early warning system to detect anomaly hot spots of agricultural production for food security analysis Agricultural Systems 2019 · 10.1016/j.agsy.2018.07.002	Operational 10-day-cadence rainfall/vegetation-index anomaly hotspot classification (four warning categories) at sub-national admin-unit level; this is the deployed real-world instantiation of the 'NDVI-threshold practice' the candidate aims to beat on lead time	Global, MODIS/ rainfall-derived, admin-unit level
Early detection of forest stress from European spruce bark beetle attack, and a new vegetation index: Normalized distance red & SWIR (NDRS) Remote Sensing of Environment 2021 · 10.1016/j.rse.2020.112240	Detects pre-visual stress using Sentinel-1+2 with accuracy 0.80-0.91 across infestation stages; found Sentinel-1 backscatter contributed only slightly beyond red/SWIR optical bands in a Random Forest	Sentinel-1+2 time series, southern Sweden spruce forest, 2018-2019

Evaluation protocol

Primary metric: lead-time-in-days, defined per flagged parcel as date(NDVI-threshold-rule alert, replicating Meroni/ASAP logic) minus date(fused SAR+optical anomaly-score threshold crossing), reported as a distribution (median, IQR) over a held-out set of parcels and growing seasons. Secondary metrics: false-alert rate per parcel-season (alerts with no corresponding NDVI-threshold crossing within a following window), and a SAR-vs-optical-vs-fused ablation reporting each channel's marginal lead-time contribution. Geographic/temporal generalisation: train the self-supervised model on one region/season's parcels only, evaluate lead time on a held-out region and/or held-out season to test whether the learned 'normal phenology' transfers. Success would be a positive median lead time (multiple days) that survives the ablation, i.e. is not attributable to optical bands alone.

Feasibility

COMPUTE

Peak memory	Estimated well under 10GB - a 3D-CAE/ConvLSTM over small spatial patches (e.g. 32x32-64x64px) per parcel time series is comparable in scale to STS-NET's reported lightweight 3D-CAE
Est. training hours (MPS)	Estimated 3-8 hours for a single-region pilot dataset (15-30GB) on Apple M4 Pro MPS backend
Fits 24 GB	Yes, comfortably

MPS-compatible ops	3D convolutions, ConvLSTM/GRU cells, and standard autoencoder losses are all supported on PyTorch MPS; no custom CUDA kernels required
Colab/Kaggle fallback	Not strictly required, but available as a buffer if the fused two-sensor time series extraction proves memory-heavy during preprocessing
Notes	Estimates are extrapolated from the architecture scale described in STS-NET's abstract and comparable published S1+S2 fusion models (e.g. Van Tricht et al. 2018 random-forest fusion, which is far lighter); not independently benchmarked in this research pass
Cash cost	Approximately RM 0 for core data and compute (Sentinel-1/2 and AI4Boundaries are free; MPS training is on the student's own MacBook Pro). A small buffer (~RM 30-50) for optional Colab Pro compute credits if the two-sensor time-series preprocessing needs more RAM than available locally. No commercial imagery dependency - the design deliberately avoids STS-NET's own use of commercial PlanetScope imagery.
Build : research	Approximately 70:30 engineering:research. The bulk of the effort is data-pipeline engineering: STAC/openEO querying, Sentinel-1 speckle filtering and orbit-based revisit alignment, Sentinel-2 cloud masking, per-parcel time-series extraction and temporal alignment across two sensors with different revisit cadences, plus the self-supervised model and the live demo dashboard. The research component is validating the lead-time claim, running the SAR-contribution ablation, and testing geographic generalisation.
Publication venue fit	IGARSS (strong fit for an applied SAR+optical fusion methods paper with a novel evaluation metric) or Remote Sensing (MDPI) for a full applied-EO paper; ICLR ML4RS workshop is also plausible given the self-supervised-anomaly-detection framing and the explicit comparison against STS-NET.

DEMO ARTIFACT

A live laptop demo replaying a held-out parcel's Sentinel-1+2 time series week-by-week: the anomaly score curve is shown rising and crossing its threshold on a specific date, plotted alongside the NDVI curve which crosses its own operational threshold N days later - the panel watches the fused-model alert fire first and sees the day-count difference displayed live, with a toggle to show the SAR-only/optical-only/fused ablation on the same parcel.

User and pain

USER SEGMENT

Primary: EU arable/horticultural farmers and the agronomists/advisory services and index-insurance providers that serve them, particularly where CAP monitoring (Sen4CAP/LPIS) already provides parcel-level infrastructure - e.g. a pilot region in France, the Netherlands, or another AI4Boundaries-covered country. Secondary/aspirational: Malaysian or Indonesian oil-palm/rice smallholder agronomists and estate managers, framed as a future deployment target, but this is weaker evidence-wise: Malaysia lacks the equivalent public parcel-level ground-truth infrastructure (LPIS/AI4Boundaries has no SEA equivalent found), and persistent cloud cover degrades the optical channel exactly where the fusion story matters most (see N1 tension). **[uncertain: final geography choice between EU and SEA not resolved by this research pass; EU is better evidenced]**

QUANTIFIED PAIN

Turvey & McLaurin (2012, Weather Climate and Society, DOI 10.1175/wcas-d-11-00059.1) found NDVI-based index insurance is highly location-dependent and unreliable without site-specific calibration across 60 US locations 1982-2003, directly evidencing that the current NDVI-threshold practice is a poor and

lagging basis for insurance decisions. The broader remote-sensing water-stress literature (Gerhards et al. 2019 review, DOI 10.3390/rs11101240) repeatedly notes that physiological water stress precedes visible canopy/NDVI symptoms, meaning yield-relevant damage is typically already underway by the time an NDVI-threshold alert fires. **[uncertain: no single hard industry figure (e.g. 'USD X million lost annually to late detection') was found in this search pass; the pain is evidenced qualitatively across multiple sources rather than by one quantified statistic]**

Key risks

1. Ground-truth/evaluation risk (highest): no independent, dated, parcel-level crop-stress-onset dataset was found publicly for the EU or SEA at the needed granularity; the self-referential NDVI-crossing proxy risks circularity (the model may just be detecting an earlier-triggering variant of the same optical signal rather than genuinely independent stress).
2. Method risk: Huo et al. (2021) found SAR contributed only marginally beyond optical bands in a closely analogous pre-visual stress detection setting (forest bark beetle); the core premise that 'SAR buys real lead time' for crops may not replicate and must be tested via ablation in week 1-2, not assumed.
3. Revisit-cadence and cloud-cover risk: Sentinel-1 (6-12 day) and Sentinel-2 (5-day nominal but cloud-degraded) have mismatched, geography-dependent effective revisit rates; a tropical SEA pilot site would face the worst cloud conditions exactly where the fusion argument is supposed to matter most (tension with N1).
4. Geography-choice risk: the EU gives better ground-truth infrastructure (AI4Boundaries, ASAP/MARS) but weakens the Singapore/Malaysia user-segment framing required to strongly satisfy H4; this needs an explicit, justified design decision, not an implicit default.
5. Data-reuse risk: STS-NET's own BSPT dataset and PlanetScope imagery are not reusable (private/commercial), so the student must assemble their own small field-verified event set from scratch, a real logistics constraint within an FYP timeline.

Kill criteria

1. Week 1: Query the Copernicus Data Space Ecosystem STAC catalogue for the candidate AOI and measure the actual per-pixel cloud-free Sentinel-2 revisit rate over a full growing season; if it is so degraded (especially for a tropical SEA site) that the optical baseline itself cannot be computed reliably, abandon that geography.
2. Week 1-2: Attempt to source at least 5-10 dated, field-verified stress events (via a local agronomist contact, a documented historical drought/flood year, or an EU national crop-condition bulletin); if zero can be obtained, decide explicitly whether the self-referential NDVI-crossing proxy is an acceptable evaluation for the panel before investing further.
3. Week 2: Run the SAR-vs-optical-vs-fused ablation on any available small sample, replicating Huo et al.'s comparison; if SAR's marginal lead-time contribution over optical alone is statistically indistinguishable from zero, reframe the pitch around fusion-robustness (e.g. cloud-gap resilience) rather than SAR-driven earliness, or abandon the SAR angle.

Open / unverified items

- labels_and_ground_truth: no independent dated parcel-level crop-stress ground-truth dataset was confirmed to exist publicly for EU or SEA; relies on a self-referential NDVI-crossing proxy pending field verification.
- user_segment: final geography choice between EU (better data infrastructure) and Malaysia/Singapore (weaker infrastructure, stronger brief alignment) is unresolved.

- quantified_pain: no single hard industry-wide loss figure was found; pain is evidenced qualitatively across multiple sources.
- total_dataset_gb and compute_feasibility: estimated by analogy to comparable published architectures (STS-NET, Van Tricht et al.), not empirically benchmarked in this pass.
- EC JRC ASAP bulk-data reuse licence terms were not independently re-verified via direct WebFetch in this pass.

Key references

STS-NET: Spatio-Temporal Stress Network for Self-Supervised Crop Stress Detection using Satellite Image Time Series

arXiv preprint · 2026 · [arXiv:2607.18791](https://arxiv.org/abs/2607.18791)

The direct named prior art the candidate must beat; confirms it uses optical-only PlanetScope imagery and reports precision, not lead time, validating the novelty delta.

Near real-time vegetation anomaly detection with MODIS NDVI: Timeliness vs. accuracy and effect of anomaly computation options

Remote Sensing of Environment · 2019 · [10.1016/j.rse.2018.11.041](https://doi.org/10.1016/j.rse.2018.11.041)

Quantifies the timeliness/accuracy tradeoff of the NDVI-anomaly practice the candidate aims to beat; source of the baseline replication logic.

ASAP: A new global early warning system to detect anomaly hot spots of agricultural production for food security analysis

Agricultural Systems · 2019 · [10.1016/j.agsy.2018.07.002](https://doi.org/10.1016/j.agsy.2018.07.002)

The real, deployed operational instantiation of the NDVI-threshold practice; grounds the 'baseline farmers and insurers use today' claim in an actual system.

Early detection of forest stress from European spruce bark beetle attack, and a new vegetation index: Normalized distance red & SWIR (NDRS)

Remote Sensing of Environment · 2021 · [10.1016/j.rse.2020.112240](https://doi.org/10.1016/j.rse.2020.112240)

Closest existing Sentinel-1+2 pre-visual-stress fusion study; its finding that SAR contributed only marginally is the key threat/risk to test early.

Potential for the Detection of Irrigation Events on Maize Plots Using Sentinel-1 Soil Moisture Products

Remote Sensing (MDPI) · 2020 · [10.3390/rs12101621](https://doi.org/10.3390/rs12101621)

Supports the biophysical premise that Sentinel-1 senses water-status changes at plot scale, underpinning the SAR-as-early-signal hypothesis.

AI4Boundaries: an open AI-ready dataset to map field boundaries with Sentinel-2 and aerial photography

Earth System Science Data · 2023 · [10.5194/essd-15-317-2023](https://doi.org/10.5194/essd-15-317-2023)

Verified free, live source of EU parcel-level geometries needed to define the per-parcel time-series extraction unit.

Applicability of the Normalized Difference Vegetation Index (NDVI) in Index-Based Crop Insurance Design

Weather, Climate, and Society · 2012 · [10.1175/wcas-d-11-00059.1](https://doi.org/10.1175/wcas-d-11-00059.1)

Quantifies the unreliability of NDVI-based index insurance, evidencing the quantified pain the candidate addresses.

Generated from the source proposal JSON. Amber [uncertain] flags are carried over verbatim from the source and mark claims that were not independently verified.